

Research Document

Algorithm Research

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1. Algorithm's

When looking into the different approaches on how music can be created based upon environmental data and algorithms, multiple approach arise to the surface. Each approach resulting in a different output, as each algorithm focusses on a different aspect regarding to the input, output and how the data is processed. To determine the most suitable algorithms for the given project context, it is essential to delve into these approaches and evaluate their applicability. In this research, the different approaches are investigated and illustrated how their interpretation of environmental data for musical composition is used.

1.1. Algorithmic Composition

Algorithmic composition is a foundational approach in computer-generated music. It relies on math, computations, or instructions to compose music either independently or collaboratively. Algorithmic composition can potentially convert environmental data into melodies, rhythms, and harmonies that mirror the changing facets of the natural world.

1.2. Sonification

Sonification, distinct from algorithmic composition, focuses on translating data into sound without adhering to traditional musical rules. With this approach, the gathered environmental data correspond to different sound features such as pitch, volume, or timbre. This creates an auditory representation, enabling listeners to identify patterns, trends, and anomalies through sound.

1.3. Data-driven Soundscapes

Data-driven soundscapes provide a holistic approach to interpreting environmental data. Instead of generating individual musical elements, this method constructs immersive auditory environments that mirror the complexity of the data. By blending ambient sounds, field recordings, and musical motifs, data-driven soundscapes offer a sensory-rich experience, offering a nuanced portrayal of environmental data.

1.4. Machine Learning/AI-based

Machine learning and artificial intelligence (AI) bring advanced computational techniques into music creation from environmental data. These algorithms can learn intricate relationships between data and music elements, resulting in compositions that adapt, evolve, and respond to environmental changes. Machine learning models like neural networks, genetic algorithms, and reinforcement learning are increasingly employed to generate music that transcends traditional rule-based approaches.

1.5. Conclusion

In this research, several algorithmic techniques for generating music from environmental data have been researched. Each of these techniques offers a distinct perspective on the relationship

between data and music. As there is wished to showcase how the different algorithms result in different output, there are three algorithms used that show a diverse range. Therefore, by combining algorithmic composition, sonification, and data-driven soundscapes, there is aimed to showcase a diverse range of musical interpretations of environmental data.